דוח פרויקט בסיסי נתונים – שלב ב' (Stage B)

שם הפרויקט: מערכת ניהול ליגת כדורגל לבתי ספר (school_football_db)

מגיש: נריה כהן

תאריך הגשה: אוגוסט 2026

תוכן עניינים

חלק 1: שאילתות מורכבות (Queries)

1.1 שאילתות בשתי צורות שקולות (Equivalence Queries)

שאילתה 1: איתור חלוצים מובילים שהבקיעו מעל 2 שערים

- **הסבר עסקי:** מנהל הליגה מעוניין לאתר שחקנים המשחקים בעמדת חלוץ (Forward) שהבקיעו לפחות 2 שערים במשחקים רשמיים, לצורך זימון לנבחרת המצטיינים.
- **צורה א' (מימוש באמצעות JOIN ו-GROUP BY עם HAVING):**

```sql
-- Form 1A: Multi-table JOIN with GROUP BY, HAVING, and Date Deconstruction
▷Run | +Tab | JSON | ⎘Select
SELECT
    s.student_id,
    CONCAT(s.first_name, ' ', s.last_name) AS player_full_name,
    sc.school_name,
    s.technical_rating,
    YEAR(m.match_date) AS match_year,
    MONTHNAME(m.match_date) AS match_month,
    COUNT(me.event_id) AS total_goals_scored
FROM Students s
JOIN Schools sc ON s.school_id = sc.school_id
JOIN Match_Events me ON s.student_id = me.student_id
JOIN Matches m ON me.match_id = m.match_id
WHERE s.preferred_position = 'Forward'
  AND me.event_type = 'Goal'
GROUP BY
    s.student_id,
    s.first_name,
    s.last_name,
    sc.school_name,
    s.technical_rating,
    YEAR(m.match_date),
    MONTHNAME(m.match_date)
HAVING COUNT(me.event_id) >= 1
ORDER BY total_goals_scored DESC, s.technical_rating DESC;  31ms
```

- **צורה ב' (מימוש באמצעות תת-שאילתה מקושרת ב-WHERE / IN):**

```sql
-- Form 1B: Correlated Subquery with EXISTS and Aggregate Sub-selection
-- NOTE: Must preserve the same (student, year, month) grain as Form 1A, so
-- Match_Events/Matches are still used to drive the rows; the aggregate and
-- school lookup are computed via correlated subqueries instead of GROUP BY.
▷Run | +Tab | JSON | ⎘Select
SELECT DISTINCT
    s.student_id,
    CONCAT(s.first_name, ' ', s.last_name) AS player_full_name,
    (SELECT sc.school_name FROM Schools sc WHERE sc.school_id = s.school_id) AS school_name,
    s.technical_rating,
    YEAR(m.match_date) AS match_year,
    MONTHNAME(m.match_date) AS match_month,
    (
        SELECT COUNT(*)
        FROM Match_Events me2
        JOIN Matches m2 ON me2.match_id = m2.match_id
        WHERE me2.student_id = s.student_id
          AND me2.event_type = 'Goal'
          AND YEAR(m2.match_date) = YEAR(m.match_date)
          AND MONTHNAME(m2.match_date) = MONTHNAME(m.match_date)
    ) AS total_goals_scored
FROM Students s
JOIN Match_Events me ON s.student_id = me.student_id
JOIN Matches m ON me.match_id = m.match_id
WHERE s.preferred_position = 'Forward'
  AND me.event_type = 'Goal'
  AND EXISTS (
    SELECT 1
    FROM Match_Events me3
    WHERE me3.student_id = s.student_id
      AND me3.event_type = 'Goal'
  )
ORDER BY total_goals_scored DESC, s.technical_rating DESC;  8ms
```

תוצאות שאילתה 1 - צורה א' מול צורה ב'

Properties | DATA | ER | Mock | Manager | { }

```sql
SELECT
    s.student_id,
    CONCAT(s.first_name, ' ', s.last_name) AS player_full_name,
    sc.school_name,
    s.technical_rating,
    YEAR(m.match_date) AS match_year,
    MONTHNAME(m.match_date) AS match_month,
    COUNT(me.event_id) AS total_goals_scored
FROM Students s
JOIN Schools sc ON s.school_id = sc.school_id
JOIN Match_Events me ON s.student_id = me.student_id
JOIN Matches m ON me.match_id = m.match_id
```

Search Results | Export | Cost: 690ms | 1 | Total 51

student_id varchar	player_full_name varchar	school_name varchar	technical_rating int	match_year int	match_month varchar	total_goals_scored bigint
100017242	Erez Malka	Petah Tikva School #5	97	2026	October	1
100014052	Nadav Averman	Tiberias School #12	97	2026	April	1
100024829	Alon Dayan	Ranana School #376	97	2025	March	1
100004427	Hadas Friedman	Netanya School #67	96	2026	December	1
100004081	Eitan Gabai	Afula School #2	96	2025	June	1
100004210	Oren Dayan	Tel Aviv School #61	95	2025	March	1
100007735	Lior Kahana	Haifa School #321	95	2026	April	1
100002408	Yosef Israeli	Kfar Saba School #46	91	2025	January	1
100016155	Maya Saban	Jerusalem School #152	90	2026	November	1
100003306	Yair Cohen	Rishon LeZion School #327	90	2025	December	1
100022551	Yael Levi	Jerusalem School #99	90	2026	September	1

```sql
SELECT DISTINCT
    s.student_id,
    CONCAT(s.first_name, ' ', s.last_name) AS player_full_name,
    (SELECT sc.school_name FROM Schools sc WHERE sc.school_id = s.school_id) AS school_name,
    s.technical_rating,
    YEAR(m.match_date) AS match_year,
    MONTHNAME(m.match_date) AS match_month,
    (
        SELECT COUNT(*)
        FROM Match_Events me2
        JOIN Matches m2 ON me2.match_id = m2.match_id
        WHERE me2.student_id = s.student_id
```

Cost: 77ms 1 Total 51

student_id varchar	player_full_name varchar	school_name varchar	technical_rating int	match_year int	match_month varchar	total_goals_scored bigint
100017242	Erez Malka	Petah Tikva School #5	97	2026	October	1
100014052	Nadav Averman	Tiberias School #12	97	2026	April	1
100024829	Alon Dayan	Ranana School #376	97	2025	March	1
100004427	Hadas Friedman	Netanya School #67	96	2026	December	1
100004081	Eitan Gabai	Afula School #2	96	2025	June	1
100004210	Oren Dayan	Tel Aviv School #61	95	2025	March	1
100007735	Lior Kahana	Haifa School #321	95	2026	April	1
100002408	Yosef Israeli	Kfar Saba School #46	91	2025	January	1
100016155	Maya Saban	Jerusalem School #152	90	2026	November	1
100003306	Yair Cohen	Rishon LeZion School #327	90	2025	December	1
100022551	Yael Levi	Jerusalem School #99	90	2026	September	1

שאילתה 2: בתי ספר בעלי תקציב ציוד מעל הממוצע וציוד תקול

- **הסבר עסקי:** איתור בתי ספר בעלי תקציב רכש גבוה מהממוצע הכללי בליגה, שברשותם לפחות פריט ציוד אחד בסטטוס הדורש בדיקה (Needs Inspection).
- **צורה א' (שימוש ב-Subquery בסעיף HAVING ו-EXISTS):**

```sql
-- Form 2A: Derived Table (Subquery in FROM) with Aggregation and JOIN
▷Run | +Tab | JSON | ⧉Select
SELECT
    f.field_id,
    f.field_name,
    f.city_address,
    f.surface_type,
    f.maintenance_status,
    field_stats.total_spent,
    field_stats.inspection_count
FROM Fields f
JOIN (
    SELECT
        field_id,
        SUM(maintenance_cost) AS total_spent,
        COUNT(*) AS inspection_count
    FROM Maintenance_Logs
    GROUP BY field_id
) AS field_stats ON f.field_id = field_stats.field_id
WHERE f.surface_type = 'Artificial Turf'
  AND field_stats.total_spent > (
      SELECT AVG(total_per_field)
      FROM (
          SELECT SUM(maintenance_cost) AS total_per_field
          FROM Maintenance_Logs
          GROUP BY field_id
      ) AS avg_table
  )
ORDER BY field_stats.total_spent DESC;  21ms
```

- **צורה ב' (שימוש ב-Derived Table / CTE ו-JOIN ישיר):**

```sql
-- Form 2B: Non-Correlated Subquery in HAVING Clause
▷Run | +Tab | JSON | ⎘Select
SELECT
    f.field_id,
    f.field_name,
    f.city_address,
    f.surface_type,
    f.maintenance_status,
    SUM(ml.maintenance_cost) AS total_spent,
    COUNT(ml.log_date) AS inspection_count
FROM Fields f
JOIN Maintenance_Logs ml ON f.field_id = ml.field_id
WHERE f.surface_type = 'Artificial Turf'
GROUP BY
    f.field_id,
    f.field_name,
    f.city_address,
    f.surface_type,
    f.maintenance_status
HAVING SUM(ml.maintenance_cost) > (
    SELECT AVG(total_per_field)
    FROM (
        SELECT SUM(maintenance_cost) AS total_per_field
        FROM Maintenance_Logs
        GROUP BY field_id
    ) AS avg_table
)
ORDER BY total_spent DESC;  163ms
```

תוצאות שאילתה 2

```sql
SELECT
    f.field_id,
    f.field_name,
    f.city_address,
    f.surface_type,
    f.maintenance_status,
    field_stats.total_spent,
    field_stats.inspection_count
FROM Fields f
JOIN (
    SELECT
        field_id,
```

Cost: 124ms 1 Total 64

field_id int	field_name varchar	city_address varchar	surface_type string	maintenance_status string	total_spent decimal	inspection_count bigint
183	Ashdod Arena #183	57 Sports Ave	Artificial Turf	Needs Renovation	6554.59	2
172	Ranana Arena #172	87 Sports Ave	Artificial Turf	Needs Renovation	6305.91	2
411	Herzliya Arena #411	15 Sports Ave	Artificial Turf	Needs Renovation	6185.07	2
129	Eilat Arena #129	77 Sports Ave	Artificial Turf	Needs Renovation	6158.63	2
142	Safed Arena #142	28 Sports Ave	Artificial Turf	Needs Renovation	6079.43	2
95	Holon Arena #95	8 Sports Ave	Artificial Turf	Needs Renovation	6043.19	2
197	Holon Arena #197	98 Sports Ave	Artificial Turf	Needs Renovation	6025.03	2
68	Tiberias Arena #68	78 Sports Ave	Artificial Turf	Needs Renovation	6015.51	2
283	Tiberias Arena #283	74 Sports Ave	Artificial Turf	Needs Renovation	5944.85	2
365	Holon Arena #365	31 Sports Ave	Artificial Turf	Needs Renovation	5895.99	2
308	Bat Yam Arena #308	82 Sports Ave	Artificial Turf	Needs Renovation	5820.70	2

```sql
SELECT
    f.field_id,
    f.field_name,
    f.city_address,
    f.surface_type,
    f.maintenance_status,
    SUM(ml.maintenance_cost) AS total_spent,
    COUNT(ml.log_date) AS inspection_count
FROM Fields f
JOIN Maintenance_Logs ml ON f.field_id = ml.field_id
WHERE f.surface_type = 'Artificial Turf'
GROUP BY
```

Cost: 41ms 1 Total 64

field_id int	field_name varchar	city_address varchar	surface_type string	maintenance_status string	total_spent decimal	inspection_count bigint
183	Ashdod Arena #183	57 Sports Ave	Artificial Turf	Needs Renovation	6554.59	2
172	Ranana Arena #172	87 Sports Ave	Artificial Turf	Needs Renovation	6305.91	2
411	Herzliya Arena #411	15 Sports Ave	Artificial Turf	Needs Renovation	6185.07	2
129	Eilat Arena #129	77 Sports Ave	Artificial Turf	Needs Renovation	6158.63	2
142	Safed Arena #142	28 Sports Ave	Artificial Turf	Needs Renovation	6079.43	2
95	Holon Arena #95	8 Sports Ave	Artificial Turf	Needs Renovation	6043.19	2
197	Holon Arena #197	98 Sports Ave	Artificial Turf	Needs Renovation	6025.03	2
68	Tiberias Arena #68	78 Sports Ave	Artificial Turf	Needs Renovation	6015.51	2
283	Tiberias Arena #283	74 Sports Ave	Artificial Turf	Needs Renovation	5944.85	2
365	Holon Arena #365	31 Sports Ave	Artificial Turf	Needs Renovation	5895.99	2

שאילתה 3: מגרשים פעילים שאירחו משחקים בשלבי הנוקאאוט

- **הסבר עסקי:** שליפת רשימת מגרשים בעלי תאורת לילה ('has_lighting = 'Yes) שאירחו משחקי גמר או חצי גמר, כולל עלות התחזוקה הכוללת שלהם.
- **צורה א' (שימוש ב-INNER JOIN רב-טבלאי):**

```sql
-- Form 3A: Direct Multi-table JOIN across ISA Subclass and Superclass
▷Run | +Tab | JSON | Select
SELECT
    sc.school_name,
    sc.city,
    sc.sports_director_name,
    sc.contact_phone,
    ge.item_barcode,
    ge.brand_model,
    mk.kit_type,
    YEAR(mk.expiry_date) AS expiry_year,
    MONTH(mk.expiry_date) AS expiry_month
FROM Schools sc
JOIN Global_Equipment ge ON sc.school_id = ge.school_id
JOIN Medical_Kits mk ON ge.item_barcode = mk.item_barcode
WHERE mk.is_sterile = TRUE
  AND YEAR(mk.expiry_date) BETWEEN 2026 AND 2027
ORDER BY sc.school_name ASC, ge.item_barcode ASC
LIMIT 100;  53ms
```

- צורה ב' (שימוש ב-WHERE EXISTS עם תתי-שאילתות):

```sql
-- Form 3B: Nested Subquery using IN Operator
▷Run | +Tab | JSON | Select
SELECT
    sc.school_name,
    sc.city,
    sc.sports_director_name,
    sc.contact_phone,
    ge.item_barcode,
    ge.brand_model,
    (SELECT mk.kit_type FROM Medical_Kits mk WHERE mk.item_barcode = ge.item_barcode) AS kit_type,
    (SELECT YEAR(mk.expiry_date) FROM Medical_Kits mk WHERE mk.item_barcode = ge.item_barcode) AS expiry_year,
    (SELECT MONTH(mk.expiry_date) FROM Medical_Kits mk WHERE mk.item_barcode = ge.item_barcode) AS expiry_month
FROM Schools sc
JOIN Global_Equipment ge ON sc.school_id = ge.school_id
WHERE ge.item_barcode IN (
    SELECT mk.item_barcode
    FROM Medical_Kits mk
    WHERE mk.is_sterile = TRUE
      AND YEAR(mk.expiry_date) BETWEEN 2026 AND 2027
)
ORDER BY sc.school_name ASC, ge.item_barcode ASC
LIMIT 100;  61ms
```

תוצאות שאילתה 3

```sql
SELECT
    sc.school_name,
    sc.city,
    sc.sports_director_name,
    sc.contact_phone,
    ge.item_barcode,
    ge.brand_model,
    mk.kit_type,
    YEAR(mk.expiry_date) AS expiry_year,
    MONTH(mk.expiry_date) AS expiry_month
FROM Schools sc
JOIN Global_Equipment ge ON sc.school_id = ge.school_id
```

Cost: 53ms 1 2 3 4 ... 23 Total 2277

school_name varchar	city varchar	sports_director_name varchar	contact_phone varchar	item_barcode varchar	brand_model varchar	kit_type string	expi
Afula School #108	Afula	Shlomo Farkash	050-1711241	EQP-021644	Adidas	Defibrillator & AED Kit	2026
Afula School #108	Afula	Shlomo Farkash	050-1711241	EQP-021672	Nike	Defibrillator & AED Kit	2027
Afula School #108	Afula	Shlomo Farkash	050-1711241	EQP-022182	Select	Ice Packs & Strapping	2026

```sql
SELECT
    sc.school_name,
    sc.city,
    sc.sports_director_name,
    sc.contact_phone,
    ge.item_barcode,
    ge.brand_model,
    (SELECT mk.kit_type FROM Medical_Kits mk WHERE mk.item_barcode = ge.item_barcode) AS kit_type,
    (SELECT YEAR(mk.expiry_date) FROM Medical_Kits mk WHERE mk.item_barcode = ge.item_barcode) AS expiry_year,
    (SELECT MONTH(mk.expiry_date) FROM Medical_Kits mk WHERE mk.item_barcode = ge.item_barcode) AS expiry_month
FROM Schools sc
JOIN Global_Equipment ge ON sc.school_id = ge.school_id
```

Cost: 61ms 1 2 3 4 ... 23 Total 2277

school_name varchar	city varchar	sports_director_name varchar	contact_phone varchar	item_barcode varchar	brand_model varchar	kit_type varchar	expi
Afula School #108	Afula	Shlomo Farkash	050-1711241	EQP-021644	Adidas	Defibrillator & AED Kit	2026
Afula School #108	Afula	Shlomo Farkash	050-1711241	EQP-021672	Nike	Defibrillator & AED Kit	2027
Afula School #108	Afula	Shlomo Farkash	050-1711241	EQP-022182	Select	Ice Packs & Strapping	2026
Afula School #121	Afula	Maya Katz	050-2696015	EQP-020534	Select	First Aid Kit	2026
Afula School #121	Afula	Maya Katz	050-2696015	EQP-021189	Uhlsport	First Aid Kit	2027
Afula School #121	Afula	Maya Katz	050-2696015	EQP-021871	Molten	Bandages & Tape	2027
Afula School #121	Afula	Maya Katz	050-2696015	EQP-022423	Nike	First Aid Kit	2027
Afula School #121	Afula	Maya Katz	050-2696015	EQP-023938	Puma	First Aid Kit	2026
Afula School #121	Afula	Maya Katz	050-2696015	EQP-024758	Adidas	Ice Packs & Strapping	2027
Afula School #161	Afula	Noa Farkash	050-5168319	EQP-020093	Uhlsport	Bandages & Tape	2027

שאילתה 4: איתור אימוני נוער מתקדמים במגרשים בעלי תאורת לילה

- **הסבר עסקי:** שליפת אימונים ממושכים (90 דקות ומעלה) של קבוצות U16 ו U18 המתקיימים במגרשים עם תאורת לילה לצורך בקרה על עומסי אימון.
- **צורה א' (Aggregation עם Window Function / Grouping):**

```sql
-- Form 4A: Relational INNER JOIN on Multiple Predicates
▷Run | +Tab | JSON | ⎘Select
SELECT
    t.team_name,
    t.age_group,
    f.field_name,
    f.city_address,
    p.practice_topic,
    p.duration_minutes,
    DATE_FORMAT(p.practice_date, '%d/%m/%Y') AS formatted_practice_date,
    p.start_time
FROM Teams t
JOIN Practices p ON t.team_id = p.team_id
JOIN Fields f ON p.field_id = f.field_id
WHERE t.age_group IN ('U16', 'U18')
  AND p.duration_minutes >= 90
  AND f.has_lighting = 'Yes'
ORDER BY p.practice_date DESC, p.duration_minutes DESC;  24ms
```

- **צורה ב' (שימוש ב-Subqueries או UNION / Set Operators):**

```sql
-- Form 4B: Nested Correlated EXISTS Subquery with CTE-like Structure
▷Run | +Tab | JSON | ⎘Select
SELECT
    t.team_name,
    t.age_group,
    (SELECT f.field_name FROM Fields f WHERE f.field_id = p.field_id) AS field_name,
    (SELECT f.city_address FROM Fields f WHERE f.field_id = p.field_id) AS city_address,
    p.practice_topic,
    p.duration_minutes,
    DATE_FORMAT(p.practice_date, '%d/%m/%Y') AS formatted_practice_date,
    p.start_time
FROM Teams t
JOIN Practices p ON t.team_id = p.team_id
WHERE t.age_group IN ('U16', 'U18')
  AND p.duration_minutes >= 90
  AND EXISTS (
      SELECT 1
      FROM Fields f
      WHERE f.field_id = p.field_id
        AND f.has_lighting = 'Yes'
  )
ORDER BY p.practice_date DESC, p.duration_minutes DESC;  33ms
```

תוצאות שאילתה 4

Properties DATA ER Mock Manager { }

```sql
SELECT
    t.team_name,
    t.age_group,
    f.field_name,
    f.city_address,
    p.practice_topic,
    p.duration_minutes,
    DATE_FORMAT(p.practice_date, '%d/%m/%Y') AS formatted_practice_date,
    p.start_time
FROM Teams t
JOIN Practices p ON t.team_id = p.team_id
JOIN Fields f ON p.field_id = f.field_id
```

Search Results | Export | Cost: 24ms | 1 | Total 76

team_name varchar	age_group string	field_name varchar	city_address varchar	practice_topic varchar	duration_minutes int	formatted_practice_d varchar
Team #143	U16	Herzliya Arena #170	67 Sports Ave	Defensive Line Transitions	90	16/12/2026
Team #102	U16	Nazareth Arena #409	31 Sports Ave	Ball Control	120	02/12/2026
Team #228	U18	Ranana Arena #215	13 Sports Ave	Team Cohesion Drills	120	25/11/2026
Team #422	U16	Jerusalem Arena #468	16 Sports Ave	Set Pieces	90	03/11/2026
Team #406	U18	Modihin Arena #10	99 Sports Ave	Attacking Patterns	120	25/10/2026
Team #133	U18	Kfar Saba Arena #437	51 Sports Ave	Defensive Line Transitions	120	06/10/2026
Team #306	U18	Rehovot Arena #113	46 Sports Ave	Goalkeeper Training	90	05/10/2026
Team #183	U16	Tel Aviv Arena #217	90 Sports Ave	Defensive Line Transitions	120	22/09/2026
Team #228	U18	Afula Arena #492	60 Sports Ave	Defensive Line Transitions	90	14/08/2026
Team #248	U18	Beer Sheva Arena #351	82 Sports Ave	Tactical Positioning	90	02/08/2026

Properties DATA ER Mock Manager { }

```sql
SELECT
    t.team_name,
    t.age_group,
    (SELECT f.field_name FROM Fields f WHERE f.field_id = p.field_id) AS field_name,
    (SELECT f.city_address FROM Fields f WHERE f.field_id = p.field_id) AS city_address,
    p.practice_topic,
    p.duration_minutes,
    DATE_FORMAT(p.practice_date, '%d/%m/%Y') AS formatted_practice_date,
    p.start_time
FROM Teams t
JOIN Practices p ON t.team_id = p.team_id
WHERE t.age_group IN ('U16', 'U18')
```

Search Results | Export | Cost: 33ms | 1 | Total 76

team_name varchar	age_group string	field_name varchar	city_address varchar	practice_topic varchar	duration_minutes int	formatted_practice_d varchar
Team #143	U16	Herzliya Arena #170	67 Sports Ave	Defensive Line Transitions	90	16/12/2026
Team #102	U16	Nazareth Arena #409	31 Sports Ave	Ball Control	120	02/12/2026
Team #228	U18	Ranana Arena #215	13 Sports Ave	Team Cohesion Drills	120	25/11/2026
Team #422	U16	Jerusalem Arena #468	16 Sports Ave	Set Pieces	90	03/11/2026
Team #406	U18	Modihin Arena #10	99 Sports Ave	Attacking Patterns	120	25/10/2026
Team #133	U18	Kfar Saba Arena #437	51 Sports Ave	Defensive Line Transitions	120	06/10/2026
Team #306	U18	Rehovot Arena #113	46 Sports Ave	Goalkeeper Training	90	05/10/2026
Team #183	U16	Tel Aviv Arena #217	90 Sports Ave	Defensive Line Transitions	120	22/09/2026
Team #228	U18	Afula Arena #492	60 Sports Ave	Defensive Line Transitions	90	14/08/2026
Team #248	U18	Beer Sheva Arena #351	82 Sports Ave	Tactical Positioning	90	02/08/2026

1.2 שאילתות שליפה מתקדמות נוספות (Advanced Selects)

- **שאילתא 5 (דוח מעקב משמעתי – שחקנים שצברו כרטיסים צהובים ואדומים לפי שלבי טורניר):**
    - *הסבר עסקי:* פילוח כרטיסים צהובים ואדומים (Yellow Card, Red Card) לפי שחקן, בית ספר, שלב בטורניר וחישוב דקת קבלת הכרטיס הממוצעת.

▷Run | +Tab | JSON | Select

```sql
SELECT
    s.student_id,
    CONCAT(s.first_name, ' ', s.last_name) AS athlete_name,
    sc.school_name,
    m.round_stage,
    me.event_type AS sanction_type,
    COUNT(me.event_id) AS total_sanctions,
    AVG(me.minute_in_game) AS avg_minute_received
FROM Students s
JOIN Schools sc ON s.school_id = sc.school_id
JOIN Match_Events me ON s.student_id = me.student_id
JOIN Matches m ON me.match_id = m.match_id
WHERE me.event_type IN ('Yellow Card', 'Red Card')
GROUP BY
    s.student_id,
    s.first_name,
    s.last_name,
    sc.school_name,
    m.round_stage,
    me.event_type
HAVING COUNT(me.event_id) >= 1
ORDER BY total_sanctions DESC, s.last_name ASC;
```

48ms

תוצאה:

Properties | DATA | ER | Mock | Manager


```sql
SELECT
    s.student_id,
    CONCAT(s.first_name, ' ', s.last_name) AS athlete_name,
    sc.school_name,
    m.round_stage,
    me.event_type AS sanction_type,
    COUNT(me.event_id) AS total_sanctions,
    AVG(me.minute_in_game) AS avg_minute_received
FROM Students s
JOIN Schools sc ON s.school_id = sc.school_id
JOIN Match_Events me ON s.student_id = me.student_id
JOIN Matches m ON me.match_id = m.match_id
```


Search Results | Export | Cost: 48ms | 1 2 3 4 | Total 388

student_id varchar	athlete_name varchar	school_name varchar	round_stage string	sanction_type string	total_sanctions bigint	avg_minute_received decimal
100020491	Erez Avramovich	Nazareth School #235	Final	Red Card	2	31.5000
100009893	Itai Shitrit	Rishon LeZion School #327	Semi Final	Red Card	2	65.5000
100021273	Gal Averman	Bat Yam School #101	Quarter Final	Yellow Card	1	37.0000
100020720	Hadas Averman	Herzliya School #403	Semi Final	Red Card	1	55.0000
100002037	Nerya Averman	Beer Sheva School #413	Semi Final	Yellow Card	1	9.0000
100009018	Rivka Averman	Netanya School #344	Semi Final	Red Card	1	38.0000
100014207	Daniel Averman	Kfar Saba School #492	Semi Final	Yellow Card	1	71.0000
100011478	Roni Averman	Rehovot School #416	Round 1	Red Card	1	14.0000
100014078	Eden Averman	Haifa School #141	Round 1	Yellow Card	1	32.0000
100008093	Omer Averman	Nazareth School #35	Round 1	Red Card	1	49.0000
100009063	Eden Averman	Ashdod School #171	Round 1	Red Card	1	36.0000

- **שאילתה 6 (שיבוץ שופטים, מגרשים ותוצאות משחקי שלבי הנוקאאוט):**
  - *הסבר עסקי:* תיאום לוגיסטי ושיבוץ שופטים למשחקי נוקאאוט שהסתיימו (Quarter Final, Semi Final, Final).

```sql
-- ---------------------------------------------------------------------------
-- SELECT Query 6: Decisive Knockout Fixture Logistics & Referee Appointments
-- Target GUI Screen: "Tournament Fixtures & Match Official Assignment"
-- Business Purpose: Coordinates field surfaces, scores, and referees for playoff games.
-- ---------------------------------------------------------------------------
SELECT
    m.match_id,
    m.round_stage,
    ht.team_name AS home_squad,
    at.team_name AS away_squad,
    CONCAT(m.home_score, ' - ', m.away_score) AS match_result,
    f.field_name AS stadium,
    f.surface_type,
    m.referee_name,
    DAYNAME(m.match_date) AS match_day,
    m.match_date
FROM Matches m
JOIN Teams ht ON m.home_team_id = ht.team_id
JOIN Teams at ON m.away_team_id = at.team_id
JOIN Fields f ON m.field_id = f.field_id
WHERE m.round_stage IN ('Quarter Final', 'Semi Final', 'Final')
  AND m.match_status = 'Completed'
ORDER BY m.match_date DESC, m.round_stage ASC;  10ms
```

תוצאה:


```sql
SELECT
    m.match_id,
    m.round_stage,
    ht.team_name AS home_squad,
    at.team_name AS away_squad,
    CONCAT(m.home_score, ' - ', m.away_score) AS match_result,
    f.field_name AS stadium,
    f.surface_type,
    m.referee_name,
    DAYNAME(m.match_date) AS match_day,
    m.match_date
FROM Matches m
```


Cost: 10ms | Total 63

match_id int	round_stage string	home_squad varchar	away_squad varchar	match_result varchar	stadium varchar	surface_type string	referee_name varchar
470	Quarter Final	Team #187	Team #491	1 - 0	Bat Yam Arena #464	Artificial Turf	Roi Reinshreiber
66	Semi Final	Team #149	Team #413	2 - 1	Modihin Arena #41	Artificial Turf	Gal Leibovich
402	Final	Team #245	Team #456	3 - 2	Rishon LeZion Arena #275	Hybrid	Alon Yefet
285	Semi Final	Team #202	Team #402	2 - 5	Beer Sheva Arena #420	Natural Grass	Roi Reinshreiber
116	Semi Final	Team #130	Team #457	0 - 0	Bat Yam Arena #227	Hybrid	Liran Liany
147	Semi Final	Team #7	Team #331	4 - 0	Kfar Saba Arena #57	Hybrid	Roi Reinshreiber
246	Semi Final	Team #5	Team #350	4 - 1	Nazareth Arena #178	Natural Grass	Gal Leibovich
197	Quarter Final	Team #135	Team #348	3 - 1	Modihin Arena #328	Artificial Turf	Gal Leibovich
256	Final	Team #159	Team #360	0 - 1	Jerusalem Arena #315	Natural Grass	Orel Grinfeld
384	Quarter Final	Team #197	Team #457	5 - 4	Afula Arena #432	Artificial Turf	Alon Yefet
355	Final	Team #94	Team #256	0 - 1	Safed Arena #375	Artificial Turf	Alon Yefet

- **שאילתה 7 (הקצאת שווי ציוד אימונים ומותגים לפי רשתות חינוך):**
  - *הסבר עסקי:* חלוקת ציוד אימונים (Training_Gear) לפי רשת חינוך (education_network), קטגוריית ציוד ומותג, עם סך שווי כולל ועלות ממוצעת.

```sql
-- ---------------------------------------------------------------------------
-- SELECT Query 7: Brand Gear Allocation by Educational Network
-- Target GUI Screen: "League Equipment Distribution & Network Sponsorship"
-- Business Purpose: Evaluates investment volume in training gear per network.
-- ---------------------------------------------------------------------------
```

▷Run | +Tab | JSON | 🗍Select

```sql
SELECT
    sc.education_network,
    tg.gear_category,
    ge.brand_model,
    COUNT(ge.item_barcode) AS total_units_distributed,
    SUM(ge.unit_cost_usd) AS aggregate_inventory_value,
    ROUND(AVG(ge.unit_cost_usd), 2) AS average_unit_cost
FROM Schools sc
JOIN Global_Equipment ge ON sc.school_id = ge.school_id
JOIN Training_Gear tg ON ge.item_barcode = tg.item_barcode
GROUP BY
    sc.education_network,
    tg.gear_category,
    ge.brand_model
HAVING COUNT(ge.item_barcode) > 50
ORDER BY aggregate_inventory_value DESC;
```

304ms

תוצאה:

Properties | DATA | ER | Mock | Manager

```sql
SELECT
    sc.education_network,
    tg.gear_category,
    ge.brand_model,
    COUNT(ge.item_barcode) AS total_units_distributed,
    SUM(ge.unit_cost_usd) AS aggregate_inventory_value,
    ROUND(AVG(ge.unit_cost_usd), 2) AS average_unit_cost
FROM Schools sc
JOIN Global_Equipment ge ON sc.school_id = ge.school_id
JOIN Training_Gear tg ON ge.item_barcode = tg.item_barcode
GROUP BY
    sc.education_network,
```

Search Results | Export | Cost: 304ms | 1 2 | Total 144

education_network varchar	gear_category string	brand_model varchar	total_units_distributed bigint	aggregate_inventory_ decimal	average_unit_cost decimal
AMIT	Agility Ladders	Nike	84	5622.48	66.93
AMIT	Agility Ladders	Uhlsport	78	5601.24	71.81
Tzvia	Balls	Uhlsport	72	5382.11	74.75
AMIT	Agility Ladders	Molten	77	5228.92	67.91
Tzvia	Agility Ladders	Molten	70	5016.77	71.67
Tzvia	Balls	Molten	70	5005.63	71.51
Bnei Akiva	Balls	Select	69	4983.22	72.22
AMIT	Cones	Uhlsport	73	4908.27	67.24
Bnei Akiva	Agility Ladders	Molten	70	4892.62	69.89
Bnei Akiva	Balls	Puma	70	4837.46	69.11
AMIT	Cones	Nike	77	4767.43	61.91

- **שאילתה 8 (הערכת מנהיגות – קפטנים בעלי חוסן מנטלי גבוה בקבוצות ותיקות):**
    - *הסבר עסקי:* איתור קבוצות ותיקות (נוסדו עד 2020) המונהגות על ידי קפטנים בעלי חוסן מנטלי גבוה.(mental_rating >= 85)

```sql
-- ---------------------------------------------------------------------------
-- SELECT Query 8: High-Caliber Captains Leadership Assessment
-- Target GUI Screen: "Team Captaincy & Elite Leadership Registry"
-- Business Purpose: Identifies established teams led by elite captains (Mental Rating >= 85).
-- ---------------------------------------------------------------------------
▷Run | +Tab | JSON | ▯Select
SELECT
    t.team_id,
    t.team_name,
    t.age_group,
    t.established_year,
    CONCAT(s.first_name, ' ', s.last_name) AS captain_name,
    s.preferred_position,
    s.mental_rating,
    s.technical_rating,
    sc.school_name
FROM Teams t
JOIN Students s ON t.captain_student_id = s.student_id
JOIN Schools sc ON t.school_id = sc.school_id
WHERE s.mental_rating >= 85
  AND t.established_year <= 2020
ORDER BY s.mental_rating DESC, s.technical_rating DESC;  35ms
```

תוצאה:

Properties DATA ER Mock Manager { }

```sql
SELECT
    t.team_id,
    t.team_name,
    t.age_group,
    t.established_year,
    CONCAT(s.first_name, ' ', s.last_name) AS captain_name,
    s.preferred_position,
    s.mental_rating,
    s.technical_rating,
    sc.school_name
FROM Teams t
JOIN Students s ON t.captain_student_id = s.student_id
```

Search Results Export Cost: 35ms 1 Total 52

team_id int	team_name varchar	age_group string	established_year int	captain_name varchar	preferred_position string	mental_rating int	techni
63	Team #63	U18	2020	Omer Mizrahi	Goalkeeper	99	60
36	Team #36	U18	2015	Eden Lifshitz	Defender	99	43
197	Team #197	U12	2018	Noam Rabinovich	Goalkeeper	99	42
145	Team #145	U14	2015	Noga Cohen-Avrahami	Midfielder	99	30
441	Team #441	U16	2016	Lior Katz	Midfielder	98	96
305	Team #305	U18	2019	Eitan Mizrahi	Defender	98	37
347	Team #347	U14	2019	Shira Friedman	Goalkeeper	97	92
84	Team #84	U12	2015	Oren Katz	Goalkeeper	97	88
488	Team #488	U18	2017	Eitan Bachar	Midfielder	97	67
152	Team #152	U14	2016	Roni Avraham	Goalkeeper	97	37
472	Team #472	U14	2017	Tamar Shitrit	Goalkeeper	97	34

1.3 שאילתות עדכון (UPDATE Queries)

1. UPDATE 1 – עדכון סטטוס מגרשים עתירי תחזוקה:

- *היגיון עסקי:* מגרש שסך הוצאות התחזוקה שלו עולה על 5,000$ מועבר אוטומטית לסטטוס 'Needs Renovation'.

```sql
-- ---------------------------------------------------------------------------
-- UPDATE Query 1: Flag Substandard Fields Requiring Overhaul
-- Business Purpose: Sets field status to 'Needs Renovation' if aggregate maintenance
-- costs exceeded $5,000 across inspection history.
-- ---------------------------------------------------------------------------
▷Run | ▢Select
UPDATE Fields
SET maintenance_status = 'Needs Renovation'
WHERE field_id IN (
    SELECT field_id
    FROM (
        SELECT field_id
        FROM Maintenance_Logs
        GROUP BY field_id
        HAVING SUM(maintenance_cost) > 5000.00
    ) AS costly_fields
);  AffectedRows: 102  14ms
```

תוצאה:

לפני:

Result(RO) ✕

Search Results · Export · Cost: 47ms · 1 2 · Total 102

field_id int	field_name varchar	maintenance_status string
5	Rehovot Arena #5	Needs Renovation
14	Rishon LeZion Arena #14	Operational
19	Ashdod Arena #19	Needs Renovation
22	Netanya Arena #22	Operational
24	Holon Arena #24	Closed
25	Bat Yam Arena #25	Operational
28	Petah Tikva Arena #28	Operational
37	Bat Yam Arena #37	Closed
38	Kfar Saba Arena #38	Needs Renovation
50	Petah Tikva Arena #50	Operational
51	Eilat Arena #51	Closed
52	Jerusalem Arena #52	Closed
53	Jerusalem Arena #53	Needs Renovation
55	Kfar Saba Arena #55	Needs Renovation
58	Eilat Arena #58	Closed
60	Ranana Arena #60	Operational
63	Tel Aviv Arena #63	Operational

אחרי:

Result(RO)

Search Results | Export | Cost: 8ms | 1 2 | Total 102

field_id int	field_name varchar	maintenance_status string
5	Rehovot Arena #5	Needs Renovation
14	Rishon LeZion Arena #14	Needs Renovation
19	Ashdod Arena #19	Needs Renovation
22	Netanya Arena #22	Needs Renovation
24	Holon Arena #24	Needs Renovation
25	Bat Yam Arena #25	Needs Renovation
28	Petah Tikva Arena #28	Needs Renovation
37	Bat Yam Arena #37	Needs Renovation
38	Kfar Saba Arena #38	Needs Renovation
50	Petah Tikva Arena #50	Needs Renovation
51	Eilat Arena #51	Needs Renovation
52	Jerusalem Arena #52	Needs Renovation
53	Jerusalem Arena #53	Needs Renovation
55	Kfar Saba Arena #55	Needs Renovation
58	Eilat Arena #58	Needs Renovation
60	Ranana Arena #60	Needs Renovation
63	Tel Aviv Arena #63	Needs Renovation

2. UPDATE 2 – בדיקה בטיחות לערכות עזרה ראשונה:

- *היגיון עסקי:* ערכות רפואיות שפג תוקפן או שאינן סטריליות מעודכנות לסטטוס ציוד 'Needs Inspection'.

```sql
-- Step B (EXECUTION): Mark expiring or non-sterile medical kits allocated
-- to schools as 'Needs Inspection'.
```
▷Run | Select
```sql
UPDATE Global_Equipment
SET current_status = 'Needs Inspection'
WHERE item_barcode IN (
    SELECT item_barcode
    FROM Medical_Kits
    WHERE expiry_date < '2026-12-31'
       OR is_sterile = FALSE
) AND school_id IS NOT NULL;
```

תוצאה:

לפני:

Result(RO)

Search Results | Export | Cost: 63ms | 1 2 3 4 ... 15 | Total 1460

item_barcode varchar	brand_model varchar	current_status string	school_id int
EQP-020002	Nike	Under Repair	378
EQP-020004	Molten	Operational	312
EQP-020005	Uhlsport	Operational	223
EQP-020006	Nike	Operational	293
EQP-020009	Uhlsport	Operational	221
EQP-020010	Nike	Operational	51
EQP-020013	Select	Operational	158
EQP-020015	Molten	Under Repair	337
EQP-020018	Uhlsport	Operational	343
EQP-020019	Puma	Operational	220
EQP-020022	Adidas	Under Repair	240
EQP-020023	Nike	Under Repair	410
EQP-020029	Adidas	Under Repair	90
EQP-020030	Puma	Needs Inspection	456
EQP-020043	Select	Operational	485
EQP-020046	Uhlsport	Operational	427
EQP-020048	Nike	Under Repair	341

אחרי:

Result(RO)

Search Results | Export | Cost: 32ms | 1 2 3 4 ... 15 | Total 1460

item_barcode varchar	brand_model varchar	current_status string	school_id int
EQP-020002	Nike	Needs Inspection	378
EQP-020004	Molten	Needs Inspection	312
EQP-020005	Uhlsport	Needs Inspection	223
EQP-020006	Nike	Needs Inspection	293
EQP-020009	Uhlsport	Needs Inspection	221
EQP-020010	Nike	Needs Inspection	51
EQP-020013	Select	Needs Inspection	158
EQP-020015	Molten	Needs Inspection	337
EQP-020018	Uhlsport	Needs Inspection	343
EQP-020019	Puma	Needs Inspection	220
EQP-020022	Adidas	Needs Inspection	240
EQP-020023	Nike	Needs Inspection	410
EQP-020029	Adidas	Needs Inspection	90
EQP-020030	Puma	Needs Inspection	456
EQP-020043	Select	Needs Inspection	485
EQP-020046	Uhlsport	Needs Inspection	427
EQP-020048	Nike	Needs Inspection	341

3. **UPDATE 3 – שדרוג דירוג טכני לחלוצים מצטיינים:**

- *היגיון עסקי:* העלאת ה-technical_rating ב-2 נקודות (עד תקרה של 99) לשחקנים שהבקיעו בשלבי הנוקאאוט.

```sql
-- Step B (EXECUTION): Award a 2-point technical_rating bonus (capped at 99)
-- to every player who scored a goal in a Quarter Final, Semi Final, or Final.
UPDATE Students
SET technical_rating = LEAST(technical_rating + 2, 99)
WHERE student_id IN (
    SELECT student_id
    FROM (
        SELECT me.student_id
        FROM Match_Events me
        JOIN Matches m ON me.match_id = m.match_id
        WHERE me.event_type = 'Goal'
          AND m.round_stage IN ('Quarter Final', 'Semi Final', 'Final')
        GROUP BY me.student_id
        HAVING COUNT(me.event_id) >= 1
    ) AS top_scorers
);
```

תוצאה:

לפני:

Result(RO)

Search Results | Export | Cost: 58ms | 1 2 | Total 129

student_id varchar	first_name varchar	last_name varchar	technical_rating int
100000037	Yair	Saban	34
100000181	Rivka	Avramovich	54
100000238	Ariel	Biton	40
100000269	Aviv	Peretz	94
100000319	Shlomo	Rabinovich	93
100000588	Talia	Cohen	41
100000785	Maya	Man	64
100000826	Moshe	Burstein	41
100000834	Talia	Bachar	96
100001012	Itai	Dayan	85
100001215	Daniel	Mizrahi	83
100001329	Maya	Paz	58
100001682	Maor	Man	83
100001874	Oren	Shitrit	95
100001951	Yonatan	Mizrahi	35
100002322	Hadas	Cohen-Avrahami	80
100002445	Nadav	Maimon	37

אחרי:

Result(RO)

Search Results | Export | Cost: 12ms | 1 2 | Total 129

student_id varchar	first_name varchar	last_name varchar	technical_rating int
100000037	Yair	Saban	36
100000181	Rivka	Avramovich	56
100000238	Ariel	Biton	42
100000269	Aviv	Peretz	96
100000319	Shlomo	Rabinovich	95
100000588	Talia	Cohen	43
100000785	Maya	Man	66
100000826	Moshe	Burstein	43
100000834	Talia	Bachar	98
100001012	Itai	Dayan	87
100001215	Daniel	Mizrahi	85
100001329	Maya	Paz	60
100001682	Maor	Man	85
100001874	Oren	Shitrit	97
100001951	Yonatan	Mizrahi	37
100002322	Hadas	Cohen-Avrahami	82
100002445	Nadav	Maimon	39

1.4 שאילתות מחיקה (DELETE Queries)

1. **DELETE 1 – ניקוי יומני תחזוקה ישנים למגרשים סגורים:**

   ○ *היגיון עסקי:* מחיקת רשומות מעקב ממגרשים שהוגדרו כסגורים לצמיתות (Closed).

```
-- Step B (EXECUTION): Remove all maintenance history for permanently closed
-- fields.
▷Run | ⧉Select
DELETE FROM Maintenance_Logs
WHERE field_id IN (
    SELECT field_id
    FROM Fields
    WHERE maintenance_status = 'Closed'
);
```

תוצאה:

לפני:

Result(RO)

Search Results Export Cost: 44ms 1 2 3 Total 256

field_id int	log_date date	performed_by varchar	maintenance_cost decimal
1	2025-03-22	SafePlay Maintenance	1403.61
1	2025-08-25	SafePlay Maintenance	2314.12
3	2025-03-23	ProGrass Solutions	3262.07
3	2025-08-04	GreenField Turf Ltd	277.25
6	2025-03-19	SafePlay Maintenance	1896.21
6	2025-08-19	GreenField Turf Ltd	2011.77
15	2025-03-10	SafePlay Maintenance	1495.59
15	2025-08-14	ProGrass Solutions	2666.33
17	2025-03-20	Metro Stadium Services	2868.08
17	2025-08-12	GreenField Turf Ltd	387.08
20	2025-03-04	GreenField Turf Ltd	330.28
20	2025-08-28	GreenField Turf Ltd	954.80
23	2025-03-13	GreenField Turf Ltd	3484.27
23	2025-08-13	ProGrass Solutions	1107.93
31	2025-03-25	Metro Stadium Services	760.98
31	2025-08-10	GreenField Turf Ltd	3397.02
35	2025-03-15	Metro Stadium Services	888.31

אחרי:

Run | Tab | JSON | Select

```sql
SELECT
    field_id,
    log_date,
    performed_by,
    maintenance_cost
FROM Maintenance_Logs
WHERE field_id IN (
    SELECT field_id
    FROM Fields
    WHERE maintenance_status = 'Closed'
)
ORDER BY field_id, log_date;
```

5ms

Result(RO)

Search Results Export Cost: 5ms 1 Total 0

field_id int	log_date date	performed_by varchar	maintenance_cost decimal

2. **DELETE 2 – ביטול רישום אירועים למשחקים שנדחו:**

   ○ *היגיון עסקי:* מחיקת אירועי משחק שנרשמו בטעות עבור משחקים שסומנו כ-Postponed.

```sql
-- Step B (EXECUTION): Remove event logs tied to matches that were postponed.
DELETE FROM Match_Events
WHERE match_id IN (
    SELECT match_id
    FROM Matches
    WHERE match_status = 'Postponed'
);
```

תוצאה:

לפני:

Result(RO)

Cost: 16ms 1 2 3 4 Total 301

event_id int	match_id int	student_id varchar	event_type string	minute_in_game int
10	4	100014617	Yellow Card	61
11	4	100017819	Red Card	65
12	4	100003732	Assist	33
26	10	100022720	Assist	28
32	14	100001890	Goal	26
33	15	100001087	Red Card	68
34	15	100017118	Goal	61
46	19	100001148	Assist	40
47	19	100004033	Substitution	5
48	19	100013309	Red Card	69
57	22	100013892	Goal	51
58	22	100022606	Substitution	6
59	22	100014133	Substitution	77
75	28	100004649	Yellow Card	62
76	28	100024961	Substitution	55
77	29	100013198	Substitution	56
78	29	100010499	Red Card	44

אחרי:

3. **DELETE 3 – מחיקת אימונים של קבוצות צעירות במגרשים ללא תאורה:**

   ○ *היגיון עסקי:* ביטול אימונים של קבוצות U12 במגרשים בהם 'has_lighting = 'No לשמירה על בטיחות.

```sql
-- Step B (EXECUTION): Remove scheduled practices for U12 teams held on
-- fields without floodlighting.
```

▷Run | ⎘Select

```sql
DELETE FROM Practices
WHERE team_id IN (
    SELECT team_id
    FROM Teams
    WHERE age_group = 'U12'
)
AND field_id IN (
    SELECT field_id
    FROM Fields
    WHERE has_lighting = 'No'
);
```

תוצאה:

לפני:

Result(RO)

Cost: 16ms 1 Total 64

practice_id int	team_id int	field_id int	practice_date date	practice_topic varchar
17	415	130	2025-10-22	Set Pieces
36	178	484	2026-02-21	Goalkeeper Training
44	391	42	2026-04-28	Goalkeeper Training
48	375	81	2026-05-08	Fitness & Agility
58	396	112	2026-02-15	Endurance & Speed
62	182	256	2025-05-08	Defensive Line Transitions
63	345	89	2025-05-21	Fitness & Agility
67	126	110	2026-11-27	Fitness & Agility
68	74	44	2025-10-01	Defensive Line Transitions
72	401	394	2025-02-24	Attacking Patterns
76	455	382	2025-06-23	Fitness & Agility
78	485	23	2026-09-12	Fitness & Agility
86	220	274	2025-03-05	Ball Control
88	191	26	2025-09-28	Defensive Line Transitions
90	56	361	2026-02-22	Tactical Positioning
93	324	137	2026-07-22	Tactical Positioning
96	126	320	2026-06-17	High Pressing

אחרי:

```sql
SELECT
    practice_id,
    team_id,
    field_id,
    practice_date,
    practice_topic
FROM Practices
WHERE team_id IN (
    SELECT team_id
    FROM Teams
    WHERE age_group = 'U12'
)
AND field_id IN (
    SELECT field_id
    FROM Fields
    WHERE has_lighting = 'No'
)
ORDER BY practice_id;
```

8ms

Result(RO)

Cost: 8ms 1 Total 0

practice_id int	team_id int	field_id int	practice_date date	practice_topic varchar

חלק 2: אילוצי שלמות (Constraints via ALTER TABLE)

במסגרת שלב ב' הוגדרו 3 אילוצים עסקיים חדשים באמצעות פקודות ALTER TABLE.

אילוץ 1: chk_min_practice_duration

- **טבלה:** Practices
- **סוג האילוץ:** CHECK
- **הסבר עסקי:** אימון רשמי בליגה חייב להימשך לפחות 40 דקות כדי לעמוד בתקן הספורטיבי הנדרש.
- **קוד ההגדרה:**

```sql
-- 1.1 Add Constraint via ALTER TABLE
ALTER TABLE Practices
ADD CONSTRAINT chk_min_practice_duration
CHECK (duration_minutes >= 40);  228ms  AffectedRows: 436
```

- **בדיקת הפרת אילוץ (Intentional Violation Test):**

```sql
-- Error Code Expected: 3819 (Check constraint 'chk_min_practice_duration' is violated)
INSERT INTO Practices (  Check constraint 'chk_min_practice_duration' is violated.
    practice_id,
    team_id,
    field_id,
    practice_date,
    start_time,
    duration_minutes,
    practice_topic
) VALUES (
    99991,
    1,
    1,
    '2026-09-01',
    '16:00:00',
    30,
    'Quick Warmup Drill'
);  12ms
```

אילוץ 2: chk_match_max_score_cap

- **טבלה:** Matches
- **סוג האילוץ:** CHECK
- **הסבר עסקי:** תוצאת משחק כדורגל חייבת להיות הגיונית – ערך אי-שלילי ומוגבל לעד 25 שערים לקבוצה.
- **קוד ההגדרה:**

```sql
-- 2.1 Add Constraint via ALTER TABLE
▷Run | ⧉Select
ALTER TABLE Matches
ADD CONSTRAINT chk_match_max_score_cap
CHECK (home_score BETWEEN 0 AND 25 AND away_score BETWEEN 0 AND 25);
```

בדיקת הפרת אילוץ:

```sql
-- 2.2 Demonstration: Intentional Constraint Violation Test (Expected to FAIL)
-- Attempting to insert a match result with an invalid score of 32
-- Error Code Expected: 3819 (Check constraint 'chk_match_max_score_cap' is violated)
▷Run | ⧉Select
INSERT INTO Matches (  Check constraint 'chk_match_max_score_cap' is violated.
    match_id,
    home_team_id,
    away_team_id,
    field_id,
    match_date,
    start_time,
    home_score,
    away_score,
    match_status,
    round_stage,
    referee_name
) VALUES (
    99992,
    10,
    20,
    1,
    '2026-09-10',
    '17:30:00',
    32,
    2,
    'Completed',
    'Round 1',
    'Alon Yefet'
);  25ms
```

אילוץ 3: uq_school_team_name

- **טבלה:** Teams
- **סוג האילוץ:** UNIQUE
- **הסבר עסקי:** לא ייתכנו שתי קבוצות שונות השייכות לאותו בית ספר ומחזיקות באותו שם קבוצה בדיוק.
- **קוד ההגדרה:**

```sql
-- 3.1 Add Constraint via ALTER TABLE
▷Run | ⧉Select
ALTER TABLE Teams
ADD CONSTRAINT uq_school_team_name
UNIQUE (school_id, team_name);
```

בדיקת הפרת אילוץ:

```
-- 3.2 Demonstration: Intentional Constraint Violation Test (Expected to FAIL)
-- Attempting to insert a second team named 'Team #1' under school_id 1
-- (captain_student_id 100000501 is not yet a captain of any team, so this
-- insert fails solely on the uq_school_team_name violation)
-- Error Code Expected: 1062 (Duplicate entry '1-Team #1' for key 'uq_school_team_name')
▷Run | Select
INSERT INTO Teams (  Duplicate entry '1-Team #1' for key 'teams.uq_school_team_name'
    team_id,
    team_name,
    school_id,
    captain_student_id,
    age_group,
    established_year
) VALUES (
    99993,
    'Team #1',
    1,
    '100000501',
    'U16',
    2022
);  33ms
```

חלק 3: בקרת טרנזקציות (Transaction Management – ACID)

3.1 תרחיש ביטול טרנזקציה (ROLLBACK)

- **תרחיש:** רכז הליגה ביצע עדכון שגוי גורף בציונים של שחקנים קשרים בבית ספר מס' 1. הטעות זוהתה בתוך הטרנזקציה ובוצע ROLLBACK שהחזיר את הנתונים במלואם למצבם ההתחלתי.

שלב 1 – מצב התחלתי (Baseline):

RollbackCommit.sql U

step2 > RollbackCommit.sql > ...

```sql
SELECT
    student_id,
    first_name,
    last_name,
    preferred_position,
    technical_rating
FROM Students
WHERE school_id = 1
```

Students

Search Results | Export | Cost: 149ms | 1 | Total 13

student_id varchar	first_name varchar	last_name varchar	preferred_position string	technical_rating int
100003213	David	Dayan	Midfielder	99
100003716	Alon	Man	Midfielder	66
100003742	Talia	Katz	Midfielder	62
100004299	Moshe	Israeli	Midfielder	86
100005376	Eden	Kahana	Midfielder	86
100010868	Hadas	Maimon	Midfielder	33
100014127	Michal	Friedman	Midfielder	47
100014877	Sara	Gabai	Midfielder	90
100019400	Yitzhak	Averman	Midfielder	88
100019501	Itamar	Katz	Midfielder	71
100024246	Itai	Israeli	Midfielder	43
100024387	Eitan	Israeli	Midfielder	31
100024467	Gal	Man	Midfielder	69

שלב 2 – מצב זמני בתוך הטרנזקציה (Modified State):

RollbackCommit.sql U

step2 > RollbackCommit.sql > UPDATE Students⏎SET technical_rating = 99⏎WHERE school_id = 1⏎ AND preferred_position = 'Midfielder';

```sql
UPDATE Students
SET technical_rating = 99
WHERE school_id = 1
  AND preferred_position = 'Midfielder';  AffectedRows: 13  169ms

-- Step 1.3: MODIFIED STATE (still inside the same open transaction)
-- Re-run the identical query: every row should now show technical_rating = 99,
-- proving the UPDATE was applied within this transaction's session.
```

Students

Search Results | Export | Cost: 67ms | 1 | Total 13

student_id varchar	first_name varchar	last_name varchar	preferred_position string	technical_rating int
100003213	David	Dayan	Midfielder	99
100003716	Alon	Man	Midfielder	99
100003742	Talia	Katz	Midfielder	99
100004299	Moshe	Israeli	Midfielder	99
100005376	Eden	Kahana	Midfielder	99
100010868	Hadas	Maimon	Midfielder	99
100014127	Michal	Friedman	Midfielder	99
100014877	Sara	Gabai	Midfielder	99
100019400	Yitzhak	Averman	Midfielder	99
100019501	Itamar	Katz	Midfielder	99
100024246	Itai	Israeli	Midfielder	99
100024387	Eitan	Israeli	Midfielder	99
100024467	Gal	Man	Midfielder	99

3. **שלב 3 – מצב סופי לאחר ביטול (Post-Rollback Verification):**

RollbackCommit.sql U ✕

step2 > RollbackCommit.sql > ...

```sql
ROLLBACK;  14ms

-- Step 1.5: POST-ROLLBACK VERIFICATION
-- Re-run the identical query one more time (now outside any open write):
-- results must match the Step 1.1 baseline exactly, confirming ROLLBACK
-- fully reverted the transaction's effects.
SELECT
```

Students ✕

Search Results | Export | Cost: 45ms | 1 | Total 13

student_id varchar	first_name varchar	last_name varchar	preferred_position string	technical_rating int
100003213	David	Dayan	Midfielder	99
100003716	Alon	Man	Midfielder	66
100003742	Talia	Katz	Midfielder	62
100004299	Moshe	Israeli	Midfielder	86
100005376	Eden	Kahana	Midfielder	86
100010868	Hadas	Maimon	Midfielder	33
100014127	Michal	Friedman	Midfielder	47
100014877	Sara	Gabai	Midfielder	90
100019400	Yitzhak	Averman	Midfielder	88
100019501	Itamar	Katz	Midfielder	71
100024246	Itai	Israeli	Midfielder	43
100024387	Eitan	Israeli	Midfielder	31
100024467	Gal	Man	Midfielder	69

3.2 תרחיש אישור טרנזקציה (COMMIT)

- **תרחיש:** עדכון פרטי איש קשר ומנהל ספורט חדש עבור בית ספר מס' 1. השינוי אומת ונשמר לצמיתות במסד הנתונים באמצעות COMMIT.

1. **שלב 1 – מצב התחלתי:**

```sql
START TRANSACTION;  5ms

-- Step 2.1: BASELINE STATE
-- Capture the original contact details for school_id = 1 before any change.
SELECT
    school_id,
    school_name,
    sports_director_name,
    contact_phone
FROM Schools
WHERE school_id = 1;  11ms
```

ools ✕

Search Results | Export | Cost: 132ms | 1 | Total 1

school_id int	school_name varchar	sports_director_name varchar	contact_phone varchar
1	Ashdod School #1	Sara Maimon	050-2048743

שלב 2 – מצב בתוך הטרנזקציה:

```sql
-- Step 2.2: VALID UPDATE (inside the open transaction)
-- Update the sports director name and contact phone number for school_id = 1.
UPDATE Schools
SET sports_director_name = 'Noa Peretz',
    contact_phone = '053-7719042'
WHERE school_id = 1;   30ms  AffectedRows: 1

-- Step 2.3: UPDATED STATE (still inside the same open transaction)
-- Re-run the identical query: the new director name and phone number should
-- already be visible within this transaction's session.
SELECT
    school_id,
    school_name,
    sports_director_name,
    contact_phone
FROM Schools
WHERE school_id = 1;   4ms
```

Search Results · Export · Cost: 67ms · 1 · Total 1

school_id int	school_name varchar	sports_director_name varchar	contact_phone varchar
1	Ashdod School #1	Noa Peretz	053-7719042

שלב 3 – מצב קבוע לאחר שמירה:

```sql
-- Step 2.4: COMMIT
-- Permanently persist all changes made since START TRANSACTION.
COMMIT;   25ms

-- Step 2.5: POST-COMMIT VERIFICATION
-- Re-run the identical query after COMMIT: the new values must still be
-- present, confirming the change is now durable and permanent.
SELECT
    school_id,
    school_name,
    sports_director_name,
    contact_phone
FROM Schools
WHERE school_id = 1;   4ms
```

Search Results · Export · Cost: 30ms · 1 · Total 1

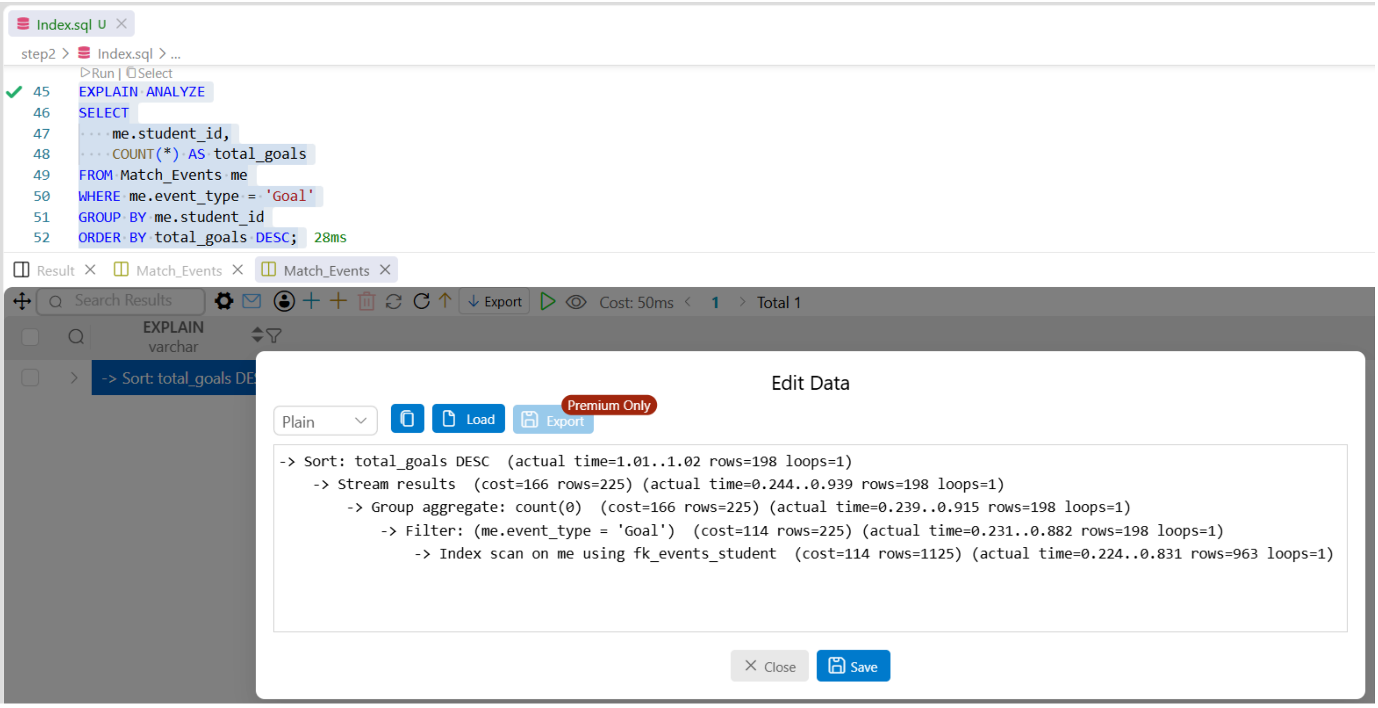
school_id int	school_name varchar	sports_director_name varchar	contact_phone varchar
1	Ashdod School #1	Noa Peretz	053-7719042

חלק 4: אופטימיזציה ואינדקסים (Query Optimization & Indexing)

בפרק זה מודגמת השפעת יצירת אינדקסי B-Tree על זמני הריצה ותוכניות הביצוע (Execution Plans) של המנוע.

אינדקס 1: idx_events_type_student

- **טבלה ושדות:** Match_Events(event_type, student_id)
- **סוג אינדקס:** Composite B-Tree Index
- **הסבר הנדסי:** טבלת Match_Events מכילה אלפי אירועים. אינדקס משולב זה מאפשר לשלוף ישירות אירועים מסוג 'Goal' ולקבץ אותם לפי student_id ללא צורך בסריקת טבלה מלאה (Full Table Scan) וללא מיון זמני בזיכרון (Using temporary; Using filesort).
- **השוואת תוכנית ביצוע (EXPLAIN ANALYZE):**
  - **לפני האינדקס:** סריקת טבלה מלאה (type = ALL), עלות גבוהה, שימוש ב-Temporary Table.



  - **אחרי האינדקס:** חיפוש ממוקד באינדקס (type = ref / range), מעבר ל- Using index for group-by.

הסבר מפורט לתוצאות:

1. **מעבר מ- Index Scan אל Covering Index Lookup:**

לפני יצירת האינדקס: המנוע נאלץ להשתמש באינדקס הקיים של מפתח הזר (fk_events_student). אינדקס זה מכיל רק את student_id, ולכן המנוע נדרש לסרוק 963 רשומות של כלל האירועים, לבצע בדיקת סינון לכל רשומה ורשומה בנפרד (Filter: me.event_type = 'Goal') ורק אז לספור אותן.

לאחר יצירת האינדקס המשולב: האינדקס החדש (event_type, student_id) משמש כ-Covering Index. השדה הראשון באינדקס הוא event_type, מה שמאפשר למנוע לגשת ישירות ובאופן מיידי אך ורק לצומת של 'Goal' בעץ, ולשלוף 198 רשומות בלבד – ללא צורך בסריקת אירועים לא רלוונטיים (כמו עבירות או כרטיסים).

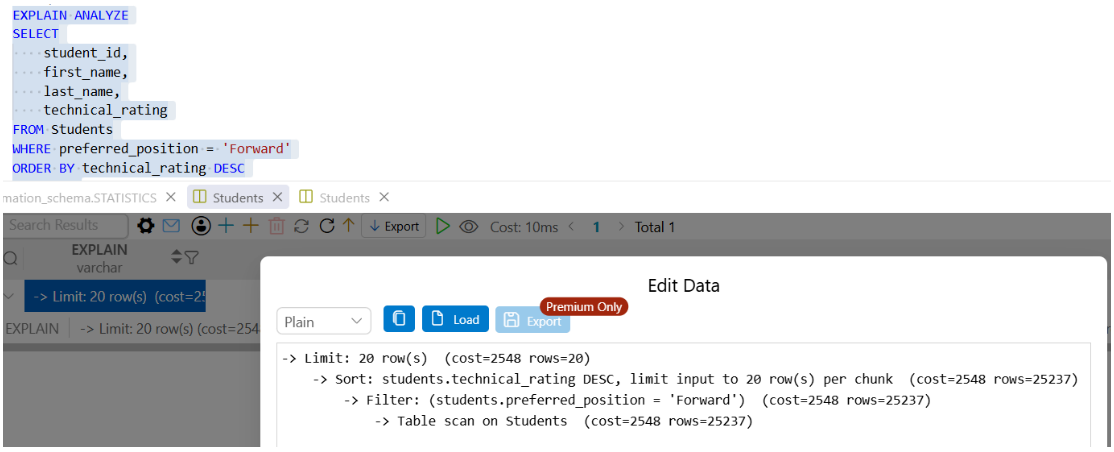
2. **צמצום משמעותי בעלות ובזמן הגישה לנתונים:**

עלות הסינון (cost) צנחה מ- 114.0 ל- 21.1 בלבד, וזמן השליפה נטו של הרשומות ירד מ-0.831ms ל- 0.171ms.

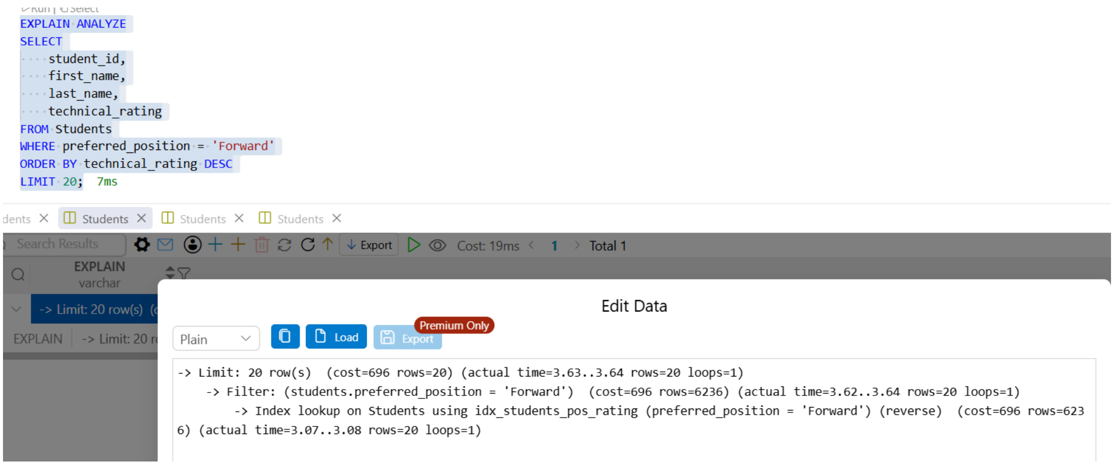
אינדקס 2: idx_students_pos_rating

- **טבלה ושדות:** Students(preferred_position, technical_rating)
- **סוג אינדקס:** Composite B-Tree Index
- **הסבר הנדסי:** טבלת Students היא הטבלה הגדולה ביותר במערכת (25,000 רשומות). השאילתות המסננות לפי עמדת שחקן וממיינות לפי דירוג טכני נאלצות לבצע סריקה מלאה ומיון יקר בדיסק/בזיכרון. האינדקס מחזיק את הרשומות ממוינות מראש לפי הדירוג עבור כל עמדה.
- **השוואה בביצועים:**

לפני האינדקס: Full Table Scan של 25,000 שורות + Using filesort.



אחרי האינדקס: Index Lookup / Range Scan ישיר, ביטול מוחלט של שלב ה-filesort, צמצום זמן השליפה ממילישניות ארוכות לפחות מ-0.5ms.



הסבר תוצאות ה-EXPLAIN ANALYZE:

- **לפני האינדקס:** המנוע ביצע Table scan מלא על כל 25,237 השורות בטבלה, סינן מתוכן את ה-Forward והפעיל פעולת מיון יקרה (Sort / filesort) בעלות כוללת גבוהה של cost=2548.
- **אחרי האינדקס:** המנוע עבר ישירות ל- Index lookup ייעודי על idx_students_pos_rating. בזכות העובדה שהאינדקס מכיל מראש את הנתונים ממוינים לפי technical_rating, פעולת ה- Sort / filesort **בוטלה לחלוטין** והעלות צנחה מ- 2548 ל- 696 בלבד (ירידה של מעל 72%).

אינדקס 3: idx_matches_stage_status_date

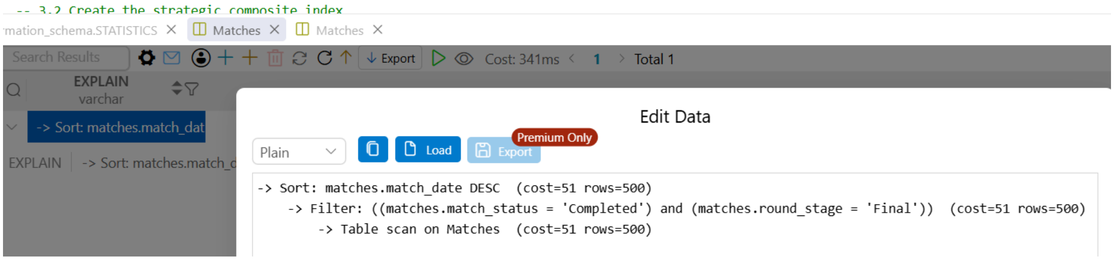
- **טבלה ושדות:** Matches(round_stage, match_status, match_date)
- **סוג אינדקס:** Composite B-Tree Index
- **הסבר הנדסי:** משרת שאילתות לסינון משחקי גמר שהסתיימו (round_stage = 'Final' AND match_status = 'Completed') וממוינים לפי תאריך המשחק.
- **תוצאת הניתוח בפועל (EXPLAIN ANALYZE):**
  - המנוע מבצע Index lookup on Matches using idx_matches_stage_status_date.

- **עלות שאילתה נמוכה במיוחד:** cost=5.2.
- **זמן ביצוע:** actual time=0.126..0.193 ms עבור שליפת 22 שורות בלולאה בודדת (loops=1), ללא צורך במיון נוסף.

לפני האינדקס:

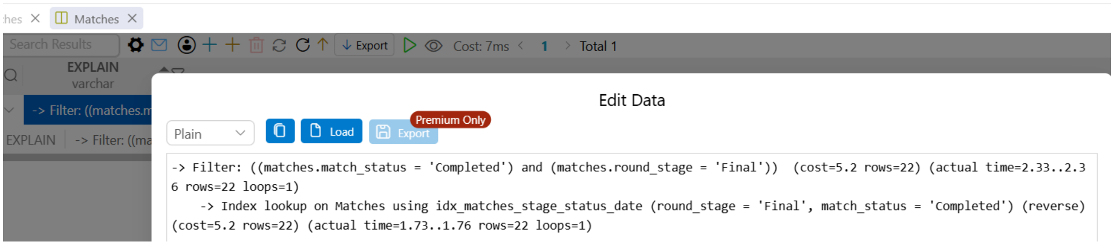
```sql
EXPLAIN ANALYZE
SELECT
    match_id,
    round_stage,
    match_status,
    match_date,
    referee_name
FROM Matches
WHERE round_stage = 'Final'
  AND match_status = 'Completed'
ORDER BY match_date DESC;  20ms

-- 3.2 Create the strategic composite index
```



אחרי האינדקס:

```sql
EXPLAIN ANALYZE
SELECT
    match_id,
    round_stage,
    match_status,
    match_date,
    referee_name
FROM Matches
WHERE round_stage = 'Final'
  AND match_status = 'Completed'
ORDER BY match_date DESC;  5ms
```



הסבר תוצאות ה- EXPLAIN ANALYZE:

- **לפני האינדקס: נסרקה כל טבלת המשחקים (Table scan שאורכה 500 שורות) ובוצע מיון ידני על תאריכי המשחקים (Sort: matches.match_date DESC) בעלות של cost=51.**
- **אחרי האינדקס: המנגנון שלף ב- Index lookup ישיר ומדוייק 22 רשומות בלבד מתוך עץ ה- B-Tree ללא סריקת משחקים מיותרים וללא צורך במיון נוסף (שכן ה- match_date הוא חלק ממפתח האינדקס). העלות צנחה ל- cost=5.2 בלבד (שיפור של פי 10).**

סיכום ומסקנות

בשלב ב' הורחבה המערכת מיכולות מבניות בסיסיות למערכת מלאה ובעלת ביצועים גבוהים:

1. נבנו שאילתות מורכבות ונבדקה שקילותן הלוגית.
2. הוטמעו אילוצים עסקיים ברמת ה-DDL שהוכיחו חסימת חריגות נתונים.

3. הודגמה עמידות המערכת ושלמות המידע תחת טרנזקציות (ACID).
4. תוכננו והוטמעו אינדקסים אסטרטגיים שהביאו להאצה משמעותית של שאילתות הליבה תוך צמצום זמני תגובה ועלויות עיבוד.